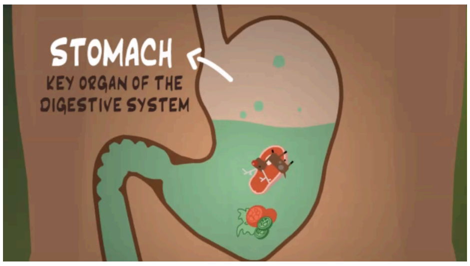
CSDS HW – Week 6

The data in this assignment is from an actual research study, but the scenario is fictional:

A health researcher is investigating how health information spreads through word-of-mouth across different media (video, pictures, text, etc.). She has prepared informative content about how to avoid stomach-aches. She is curious which **media format** to use, or avoid, in order to encourage people to **share** such health-related information.



So she prepares the following four alternative media that share the same informational content:

(1) *video [animation + audio] (click to see video): A fully animated video with audio narration*

(2) *video [pictures + audio]: Video of sequence of still pictures (not-animated) with audio narration*

(3) *webpage [pictures + text]: Static web page with still pictures (no video) and accompanying text narration (no audio)*

(4) *webpage [text only]: Static web page of text narration (no audio) but no pictures*

Our researcher runs an experiment where each of these four alternative media is shown to a different group of randomly assigned people. Afterwards, viewers were surveyed about their thoughts, including a question (labeled *`INTEND.0`*) about their intention to share what they had seen with others:

`[INTEND.0]: I intend to share the information I saw with others.`

(answered on 7 point scale: 1=strongly disagree; 4=neutral; 7=strongly agree)

You may find the researcher's data in a ZIP file containing four CSV files named: *pls-media[1-4].csv*
(note: the number in the filename corresponds to the type of media listed above)

Note: She wants to conduct her tests at 95% confidence (5% significance).

Question 1) Let's **explore and describe** the data and develop some early intuitive thoughts:

a. What are the *means* of viewers' intentions to share (INTEND.0) on each of the four media types?
b. Visualize the *distribution and mean* of intention to share, across all four media.
   (Your choice of data visualization; Try to put them all on the same plot and make it look sensible)
c. From the visualization alone, do you feel that media type makes a difference on intention to share?

Question 2) Let's try **traditional one-way ANOVA**:

a. State the null and alternative hypotheses when comparing `INTEND.0` across four groups in ANOVA
b. Let's compute the F-statistic ourselves:
   i. Show the code and results of computing MSTR, MSE, and F
   ii. Compute the p-value of F, from the null F-distribution; is the F-value significant?
   If so, state your conclusion for the hypotheses.
c. Conduct the same one-way ANOVA using the `aov()` function in R – confirm that you got similar results.
d. Regardless of your conclusions, conduct a post-hoc Tukey test (feel free to use the `TukeyHSD()` function included in base R) to see if any *pairs of media have significantly different means* – what do you find?
e. Do you feel the classic requirements of one-way ANOVA were met?
   (Feel free to use any combination of methods we saw in class or any analysis we haven't covered)

See next page for question 3

Question 3) Let's use the **non-parametric Kruskal Wallis** test:

a. State the null and alternative hypotheses
b. Let's compute an approximate Kruskal Wallis H ourselves (use the formula we saw in class or another formula might have found at a reputable website/book):
    i. Show the code and results of computing H
    ii. Compute the p-value of H, from the null chi-square distribution; is the H value significant at 5% significance? If so, state your conclusion of the hypotheses.
c. Conduct the same test using the `kruskal.test()` function in R – confirm that you got similar results.
d. Regardless of your conclusions, conduct a post-hoc Dunn test (feel free to use the `dunnTest()` function from the `FSA` package) to see if the values of any *pairs of media are significantly different* – what are your conclusions?